

Retail Business Performance & Profitability Analysis

Elevate Labs — Data Analyst Internship Project Report

1. Introduction

This project analyzes transactional retail data to identify profit-draining product categories, evaluate the impact of discounting strategies on profitability, and examine performance across regions and customer segments. The goal is to derive actionable, data-backed recommendations that a retail business could use to improve overall profitability.

2. Abstract

The Superstore dataset (9,994 transaction records) was analyzed using SQL, Python, and Tableau. Category, sub-category, region, and discount-wise profitability patterns were examined to uncover where the business gains and loses money. An interactive Tableau dashboard was built to visualize these findings, and the analysis revealed a clear relationship between high discount levels and reduced (or negative) profit margins.

3. Tools Used

- Python (Pandas, Seaborn, Matplotlib) — data cleaning, correlation analysis, and visualization
- SQL (SQLite) — category, region, and segment-wise profitability queries
- Tableau Public — interactive dashboard with filters and KPI visualizations

4. Steps Involved in Building the Project

1. Data Cleaning: Removed 17 duplicate rows; no missing values were found in the dataset.
2. SQL Analysis: Calculated profit margins by category, sub-category, region, and segment using aggregate SQL queries.
3. Python Analysis: Built a correlation matrix (Discount, Profit, Quantity, Sales) and grouped orders into discount bands to study their effect on average profit.
4. Tableau Dashboard: Built six visualizations — profit by sub-category, profit by region, average profit by discount bucket, and KPI cards for total sales, total profit, and overall profit margin — combined into one interactive dashboard with Region and Category filters.

5. Key Findings

- **Loss-making sub-category:** The [Tables] sub-category shows a negative profit despite notable sales volume, making it the primary profit drain.
- **Regional performance gap:** The [West] region delivers the highest profit, while the [Central] region lags behind, partly due to a higher average discount rate.
- **Discount impact:** Orders with discounts above 30–40% show average profit turning negative, while orders with no discount remain solidly profitable.
- **Correlation evidence:** The correlation between Discount and Profit is negative ($[-0.22]$), statistically confirming that heavier discounting directly erodes profitability.

6. Conclusion & Recommendations

The analysis shows that profitability issues are concentrated in specific sub-categories and are strongly linked to excessive discounting rather than low sales volume. It is recommended that the business cap discounts at around 20% for historically loss-making sub-categories, review pricing and discount policy in the underperforming region, and reallocate marketing focus toward high-margin categories. These steps could meaningfully improve overall profit margin without reducing sales volume significantly.
